Harrison W. Smith

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EDUCATION

Doctor of Philosophy in Environmental Dynamics
May 2026
Environmental Dynamics Program (ENDY), University of Arkansas**,** Fayetteville, AR
Dissertation Topic: *Geospatial science and the changing environment: Applied methods for sustainable agriculture and landscape management*

Master of Science in Environment and Sustainability
Apr 2019
School for Environment and Sustainability (SEAS), University of Michigan, Ann Arbor, MI.
Specializations in geospatial data science and conservation ecology.
Master's Thesis: *Evaluating multiple sensors for mapping cropped area of smallholder farms in the eastern Indo-Gangetic Plains*

Bachelor of Science in Biological Sciences (*Cum Laude)*; Minor in Spanish.
May 2016
Fulbright School of Arts and Sciences, University of Arkansas
Study abroad at the School for International Training (Ecuador) and the School for Field Studies (Peru)
Honors Research Thesis**:** *Relationships between nutrients, periphyton abundance, and benthic macroinvertebrate scrapers in Ozark Highland streams*

RESEARCH & PROFESSIONAL EXPERIENCE

ORISE Postdoctoral Fellow, USDA-ARS Poultry Production and Product Safety Research Unit
(May 2026 – Present)

- Lead UAV hyperspectral and multispectral phenotyping of soybean field trials, building Python pipelines for spectral extraction and vegetation masking
- Design a precision poultry litter study across six constructed watersheds, combining GPS-guided litter application, UAV phenotyping, and geostatistical soil sampling
- Model alfalfa and cotton suitability on Tribal lands in Arizona using MaxEnt with spatial block cross-validation
- Mentor a master's student, undergraduate interns, and work-study students in UAV data collection, spectral processing, and machine learning in R and Python

NIFA Predoctoral Fellow, Environmental Dynamics, University of Arkansas
(June 2023 – May 2026)

- Awarded a $180,000 USDA National Institute of Food and Agriculture (NIFA) predoctoral fellowship for the project "Where Can We Grow? Machine Learning to Predict Climate Impacts on U.S. Corn and Soybean Suitability"
- Built interpretable machine learning models (Random Forest, SHAP) from yield monitor and satellite data to identify environmental drivers of U.S. corn and soybean yield; published in Scientific Reports
- Predict U.S. corn yield from Landsat imagery using Random Forest, SHAP, and leave-one-site-out cross-validation on the USDA SCINet Ceres HPC cluster (manuscript in review)
- Award featured in University of Arkansas News: https://news.uark.edu/articles/64574/environmental-dynamics-student-earns-prestigious-180-000-fellowship

Graduate Research Fellow, Environmental Dynamics, University of Arkansas (Aug 2021 – June 2023)

- Conducted interdisciplinary research in soil science and sustainable agriculture
- Modeled crop suitability on Quapaw Tribal Lands for a smartphone app serving Tribal producers
- Mapped post-reclamation vegetation recovery at the Tar Creek Superfund Site using satellite time series
- Awarded the University of Arkansas Doctoral Academy Fellowship ($48,000 over four years)

Biological Sciences Technician, Dale Bumpers Small Farms Research Center, USDA-ARS (Mar 2021 – Aug 2021)

- Established the center's Unmanned Aerial Systems (UAS) program, including pilot training and flight operations
- Ran rainfall simulation experiments and designed precision agriculture figures for a 3-D touchable exhibit at the United States Botanic Garden

Project/Program Specialist, University of Arkansas Division of Agriculture (Jul 2020 – Mar 2021)

- Collaborated with USDA-ARS and University of Arkansas scientists on grant-funded sustainable agriculture research
- Applied ground penetrating radar to characterize subsurface soil properties and vegetation in a silvopasture system

Research Associate, Sustainable Agriculture & Remote Sensing, University of Michigan (Sep 2019 – May 2020)

- Systematically reviewed the literature on global agricultural adaptation to climate change and statistically analyzed sustainable intensification technologies for smallholder farmers in India
- Tested multi-sensor data fusion and semi-automated, phenology-based classification to map cover crops and sowing dates
- Built an agent-based model of agricultural adaptation to climate change in NetLogo for an educational case study: https://www.learngala.com/cases/growing-into-an-uncertain-future

Remote Sensing Intern, Michigan Tech Research Institute (May 2019 – Aug 2019)

- Automated pipelines for Synthetic Aperture Radar (SAR) processing using Python and PCI Geomatica
- Improved classification of surface water extent, flooded vegetation, and invasive species through manual calibration and rule set generation
- Conducted intensive fieldwork in the Great Lakes Basin and Northwest Territories of Canada

GIS and Mapping Intern, The Farm at St. Joe's (May 2018 – Sep 2018)

- Supported farm operations, including bed preparation, maintenance, harvest, packaging, and storage of vegetable crops
- Collected field geospatial data with mobile GIS and GPS and compiled a database from public geospatial layers and historical documents
- Developed user-friendly print and web trail maps with an interdisciplinary team

PUBLICATIONS

1. **Smith, H. W.**, Heffernan, C. J., Ashworth, A. J., Nalley, L. L., Bullock, D. S., Tullis, J., & Owens, P. R., *2026*. Harvesting insights: interpretable machine learning to understand environmental drivers of US maize and soybean yield. *Scientific Reports*. https://doi.org/10.1038/s41598-026-38724-z
2. Pillai, N., Virupakshaiah, A., **Smith, H.W.**, Ashworth, A.J., Gowda, P., Owens, P.R., Rivers, A.R., Nanduri, B. and Ramkumar, M., *2025*. Policy-driven transfer learning in resource-limited animal monitoring. *arXiv preprint arXiv:2509.10995*. https://doi.org/10.48550/arXiv.2509.10995
3. Ashworth, A.J., Avila, A., **Smith, H.W.**, Winzeler, T.E., Owens, P.R., Flynn, C., O'Brien, P., Philipp, D. and Su, J., *2024*. Predicting spatiotemporal patterns of productivity and grazing from

multispectral data using neural network analysis based on system complexity. *Agrosystems, Geosciences & Environment*. https://doi.org/10.1002/agg2.20571

4. **Smith, H.W.**, Ashworth, A.J., King, S., Kreman, C., Miller, D.M., Nalley, L.L., and Owens, P.R., *2024*. Remote sensing reveals trends in vegetative recovery and land cover transformation post-reclamation at Tar Creek Superfund Site. *Discover Geoscience*. https://doi.org/10.1007/s44288-024-00057-7
5. **Smith, H.W.**, Ashworth, A.J., Nalley, L.L., Schmidt, A., Turmel, M.S., and Owens, P.R., *2023.* Boundary line analysis and machine learning models to identify critical soil values for major crops in Guatemala. *Agronomy Journal*. https://doi.org/10.1002/agj2.21412
6. Ylagan, S., Brye, K.R., Ashworth, A.J., Owens, P.R., **Smith, H.W.**, Poncet, A.M., Sauer, T.J., Thomas, A.L., and Philipp, D., *2023*. Relationships among apparent electrical conductivity and plant and terrain data in an agroforestry system in the Ozark Highlands. *Agrosystems, Geosciences, & Environment.* http://dx.doi.org/10.1002/agg2.20414
7. **Smith, H.W.**, Owens, P.R., and Ashworth, A.J., 2022. Applications and analytical methods of ground penetrating radar for soil characterization in a silvopastoral system. *Journal of Environmental and Engineering Geophysics*. https://doi.org/10.32389/JEEG22-001
8. Ylagan, S., Brye, K.R., Ashworth, A.J., Owens, P.R., **Smith, H.W.**, Poncet, A.M., 2022. Using Apparent Electrical Conductivity to Delineate Field Variation in an Agroforestry System in the Ozark Highlands. *Remote* Sensing. DOI: https://doi.org/10.3390/rs14225777
9. **Smith, H.W.**, Ashworth, A.J., and Owens, P.R., 2022. GIS-based multicriteria evaluation of crop and soil suitability for optimized production on U.S. Tribal Lands. *Agriculture.* DOI: https://doi.org/10.3390/agriculture12091307

MEETING AND WORKSHOP ATTENDANCE

1. USDA Multistate Project S1069: Research and Extension for Unmanned Aircraft Systems (UAS) Applications in U.S. Agriculture and Natural Resources. 2023 Annual Meeting. Virginia Tech, June 1-2, 2023.
2. Advanced Topics in Deep Learning. USDA ARS SCINet Initiative workshop. Virtual Event, May 27-28, 2021.
3. Intro to Image Processing, Classical ML, and Deep Learning. USDA ARS SCINet Initiative workshop. Virtual Event, May 24-26, 2021.

PRESENTATIONS

1. O'Neill, C., L. Nalley, J. Tack, K. Ala-Kokko, S. Maali, E. Eli, H. De Steur and **H. Smith.** Nonlinear Heat Effects in South African Maize. Paper presented to the Agricultural and Applied Economics Association (AAEA) conference. July 26-29, 2026. Kansas City, Missouri.
2. **Smith, H.W.**, Li, A., Ashworth, A.J., Bullock, D.S., Harnessing multi-temporal satellite imagery and historical yield monitor data to predict U.S. maize and soybean yield with machine learning. *AGU 2025*. New Orleans, LA, December 15-19, 2025.
3. **Smith, H.W.**, Heffernan, C.J., Ashworth, A.J., Bullock, D., Nalley, L.L., Tullis, J., Owens, P.R. Harvesting insights: Yield monitor and machine learning to understand environmental drivers of U.S. maize and soybean yield. *CANVAS 2025*. Salt Lake City, UT, November 9, 2025.
4. **Smith, H.W.**, Heffernan, C.J., Ashworth, A.J., Bullock, D., Nalley, L.L., Tullis, J., Owens, P.R. Harvesting Insights: Machine learning to predict U.S. maize and soybean yields. *Crop Soil and Environmental Sciences Departmental Seminar*. University of Arkansas, Fayetteville, AR, February 3, 2025.
5. **Smith, H.W.**, Ashworth, A.J., Bullock, D., Kharel, T.P., Nalley, L.L., Owens, P.R. Where can we grow? Machine learning to predict the future of U.S. corn. *Three Minute Thesis Competition Final Round.* University of Arkansas, Fayetteville, AR, November 23, 2024.

6. **Smith, H.W.**, Ashworth, A.J., Bullock, D., Kharel, T.P., Nalley, L.L., Owens, P.R. Understanding the drivers of maize and soybean yields in U.S. agricultural landscapes using machine learning. *International Association of Landscape Ecologists, North America Annual Meeting*, Oklahoma City, OK, April 2, 2024.
7. **Smith, H.W.**, Ashworth, A.J., Bullock, D., Kharel, T.P., Nalley, L.L., Owens, P.R. Machine learning to understand and predict soil, climate, and terrain effects on suitability of U.S. maize and soybean crops. *American Geophysical Union Annual Meeting*, San Francisco, CA, December 14, 2023.
8. **Smith, H.W.**, Ashworth A.J., King, S., Kreman, C., Owens, P.R., Nalley, L.L., Miller, D. Timeseries Satellite Imagery Enables Monitoring of Vegetation Recovery at the Tar Creek Superfund Site. *ASA-CSSA-SSSA International Annual Meeting,* St. Lois, MO, October 30, 2023. Awarded 3rd place best presentation in Airborne and Satellite Remote Sensing category.
9. **Smith, H.W.**, Ashworth A.J., Nalley, L.L., Schmidt, A., Turmel, M.S., Owens, P.R., Optimizing Yields for Major Crops in Guatemala Using Boundary Line Analysis and Machine Learning Models. *ASA-CSSA-SSSA International Annual Meeting*, St. Lois, MO, October 30, 2023.
10. **Smith, H.W.**, Introduction to boundary line analysis (international workshop, delivered in Spanish). *Catholic Relief Services*. Invited Presentation. October 12, 2023.
11. **Smith, H. W.** High resolution mapping of agroforestry productivity using multispectral and hyperspectral data. *USDA Multistate Project S1069: Research and Extension for Unmanned Aircraft Systems (UAS) Applications in U.S. Agriculture and Natural Resources*. 2023 Annual Meeting. Virginia Tech, June 1-2, 2023.
12. **Smith, H.W.,** Ashworth A.J., Nalley, L.L., Schmidt, A., Turmel, M.S., Owens, P.R., Boundary line analysis can improve nutrient management in Guatemalan maize, common bean, and coffee farms. *Graduate Student Professional Congress Colloquium*, University of Arkansas, Fayetteville, AR, April 17, 2023.
13. **Smith, H. W.** Monitoring and Assessment of Reclamation at Tar Creek Superfund Site Using Timeseries Satellite Imagery. *Tribal Land and Environment Forum (TLEF) 2022*, Milwaukee, WI, August 8-11, 2022.
14. **Smith, H. W.** Geospatial analytics for enhanced soil management. *National University of Sciences and Technology, Pakistan*. Invited presentation. February 28, 2022.
15. **Smith, H. W.** Ground penetrating radar for soil characterization in an experimental agroforestry site using qualitative and quantitative techniques. *Environmental and Engineering Geophysical Society Talk About Geophysics (TAG) Webinar*. Invited presentation. December 8, 2021.
16. **Smith, H. W.** Ground penetrating radar for subsurface soil characterization and analysis**.** *Environmental and Engineering Geophysical Society/Society of Exploration Geophysicists Symposium: Application of Proximal and Remote Sensing Technologies for Soil Investigations*, Virtual Event, August 16 – 19, 2021. Awarded "1st Place Best Student Presentation.
17. **Smith, H. W.** Soil characterization with ground penetrating radar at an experimental agroforestry site**.** *North American Agroforestry Conference*. June 28 – July 2, 2021.

TEACHING EXPERIENCE

Lesson Planner, USDA-ARS Three Sisters Program, (Southern University Lab High School) 2024

- Developed curriculum and materials covering UAVs for assessments, data gathering and management in agricultural applications
- Educated 122 middle school students and staff on drone and remote sensing technology in agriculture

Graduate Student Instructor, Ecology: Concepts and Applications (University of Michigan) 2018

- Taught graduate-level ecology lab and coordinated weekly field trips to a diversity of local habitats
- Used hands-on approach to teach students ecological field methods and experimental design

Graduate Student Instructor, Natural Resource Statistics (University of Michigan) 2019

- Instructed graduate-level lab section on statistical methods using R project software
- Assisted students with de-bugging R code and clarified statistical concepts during weekly office hours

Invited Guest Lecturer, Remote Sensing of Environment (University of Michigan) 2019 - 2022

- Invited to deliver annual talk on the Google Earth Engine platform to graduate class in Remote Sensing
- Meet with students 1-on-1 to provide technical support for use of Google Earth Engine

LEADERSHIP AND MENTORING EXPERIENCE

- Undergraduate Mentorship (University of Arkansas)
    - Christopher Heffernan – University of Arkansas Computer Science Major
    - Andrew Li – USDA Wallace Carver Fellow
- Treasurer, Environmental Dynamics Student Association (University of Arkansas)
- At-Large Representative, Graduate Professional Student Congress (University of Arkansas)
- Team Lead, Pollinator Partnership Northwest Arkansas (Project Wingspan)
- Careers Chair, School for Environment and Sustainability Student Government (University of Michigan)
- Track Leader, School for Environment and Sustainability Environmental Informatics track (University of Michigan)
- Board Member, Celebrating Solomon Scholarship (Arkansas Community Foundation)

AWARDS AND HONORS

- **Rockey FFAR Fellowship**, Foundation for Food and Agriculture. 2024-2027.
- **AFRI Predoctoral Fellowship**, National Institute of Food and Agriculture. 2023-2026.
- **Doctoral Academy Fellowship**, University of Arkansas. 2021 – 2025.
- **Gerald O. Mott Meritorious Graduate Student Award in Crop Science**, Crop Science Society of America. 2024.
- **IALE-NA Student Travel Award**, International Association of Landscape Ecology - North America Chapter. 2024.
- **3rd Place Best Presentation**, Airborne and Satellite Remote Sensing category, ASA-CSSA-SSSA International Annual Meeting. 2023.
- **1st Place Best Student Presentation.** Environmental and Engineering Geophysical Society/Society of Exploration Geophysicists Symposium: Application of Proximal and Remote Sensing Technologies for Soil Investigations. 2021.
- **Honorable Mention.** Three Minute Thesis Competition, University of Arkansas: Ground Penetrating Radar for Soil Characterization in an Agroforestry System. 2022.
- **Outstanding Graduate Student Instructor**. School for Environment and Sustainability at University of Michigan. 2019.
- **Edna Bailey Sussman Award for Environmental Internships**, Edna Bailey Sussman Fund. 2018.
- **Governor's Distinguished Scholarship**, Arkansas Department of Higher Education. 2012 – 2016.
- **Chancellor's Scholarship**, University of Arkansas. 2012 – 2016.

GRANTS

1. AFRI Predoctoral Fellowship, National Institute of Food and Agriculture (NIFA). "Where can we grow? Machine learning to predict climate impacts on U.S. corn and soybean suitability" (PI). 2023-2026. $180,000.

2. FFAR Predoctoral Fellowship, University of Arkansas (PI). 2024-2027. $90,000.
3. Doctoral Academy Fellowship, University of Arkansas (PI). 2021-2025. $48,000.
4. University of Arkansas Doctoral Student Travel Grant for "ASA-CSSA-SSSA International Annual Meeting", October 30, 2023" (PI). October 29 - November 1, 2023. $750.
5. University of Arkansas Doctoral Student Travel Grant for "Tribal Land and Environment Forum (TLEF) 2022" (PI). August 8-11, 2022. $1,000.
6. USDA-ARS Innovation Fund (2022). "Quantifying air and water quality benefits of improved poultry manure management practices" (Co-PI). $25,000.

COMMUNITY SERVICE

Board President, Tri Cycle Farms, Fayetteville, AR

- Assists the board of Tri Cycle in management of the farm and consults with board members regarding ways to enhance food recovery and redistribution programs.

Volunteer Technician, NWA Black Heritage East Mountain Cemetery Project

- In partnership with the Arkansas Archeological Survey, a group of interested students and community members used proximal sensing techniques (electrical resistivity, ground penetrating radar) to search for unmarked graves at East Mountain Cemetery
- Information on the project can be found at: https://nwablackheritage.org/cemeteries

YouthMappers at UArk, University of Arkansas

- Attend regular meetings, workshops, and events to connect with fellow geospatial researchers and practitioners
- Contributes to volunteer and citizen science opportunities such as projects for disaster relief in Humanitarian OpenStreetMap: https://www.hotosm.org/

PROFESSIONAL MEMBERSHIPS

- American Society of Agronomy
- Crop Science Society of America
- Soil Science Society of America
- American Geophysical Union
- Ecological Society of America
- International Association of Landscape Ecology